1 Atmospheric Delay Modelling: Troposphere and Ionosphere

The undifferenced code and phase equations, the code observation and the phase observation of the Foundations document, each contain two propagation terms that are computed from *models* rather than estimated directly: the slant tropospheric delay T_r^i and the first-order slant ionospheric delay $\mu_s I_r^i$. The troposphere has an a-priori *zenith hydrostatic* part, which is computed and removed before the filter sees the measurement. The small *zenith wet* residual and two horizontal gradients are estimated (the state vector, the ZWD mapping). In the uncombined parameterisation the ionosphere is one slant state per satellite (dynamic indexing). The models in this chapter supply its a-priori seed, an optional pseudo-observation that ties the relative slants to a Global Ionosphere Map, and the small second- and third-order corrections that the $1/f^2$ scaling does not cover. The network producer also co-estimates its own spherical-harmonic VTEC field on one thin shell, or optionally two (Section 1.2.4), so it needs no external GIM. The column layout of that field in the network normal equations is in the network mode chapter. This document derives the mapping functions, the pressure reductions, the thin-shell geometry, the geomagnetic field, and the sign with which each term enters the modelled range.

1.1 The slant tropospheric delay

The neutral atmosphere is non-dispersive at GNSS frequencies. Its delay is the same on code and phase, and no ionosphere-free or geometry-free combination removes it. The standard decomposition splits the slant delay into a hydrostatic ("dry") and a wet zenith delay, each carried to the line of sight by its own elevation mapping function, plus an azimuthal gradient [7]:

$$\boxed{T_r^i = m_{\mathrm{h}}(\varepsilon)\,\mathrm{ZHD} + m_{\mathrm{w}}(\varepsilon)\,\mathrm{ZWD} + m_{\mathrm{g}}(\varepsilon)\big(G_{\mathrm{N}}\cos\alpha + G_{\mathrm{E}}\sin\alpha\big).} \tag{1}$$

ZHD is supplied a priori and removed from the prefit residual. ZWD, G_{N} and G_{E} are estimated states whose Jacobian columns are the mapping coefficients $m_{\mathrm{w}}(\varepsilon)$ and $m_{\mathrm{g}}(\varepsilon)\{\cos\alpha, \sin\alpha\}$ (`src/core/measurement/MeasurementModel.cpp`; the ZWD mapping). The rest of this subsection derives each factor.

1.1.1 Saastamoinen zenith hydrostatic delay

The total surface pressure determines the zenith hydrostatic delay to better than a millimetre, through the Saastamoinen reduction [6] as coded in `src/core/atmosphere/Troposphere.cpp`:

$$\boxed{\mathrm{ZHD} = \frac{0.0022768\ P}{1 - 0.00266\cos 2\phi - 0.00028\,h_{\mathrm{km}}},} \tag{2}$$

with P the total pressure in millibar at the antenna, ϕ the geodetic latitude, and $h_{\mathrm{km}} = h/1000$ the ellipsoidal height in kilometres (`Troposphere::computeSaastamoinenZHD`, denominator `f_lat`). The coefficient 0.0022768 m mbar^{-1} combines the refractivity constant $k_1 = 77.604$ K mbar^{-1} with the dry-gas constant and standard gravity. The denominator $f(\phi, h)$ accounts for the variation of gravity with latitude and height.

When neither a surface met sensor nor the GPT3 grid (below) is available, P is taken from a standard-atmosphere barometric reduction of a configured sea-level pressure P_0 and temperature T_0

(`Troposphere.cpp`, `pressureFromHeightStdAtm`):

$$P(h) = P_0 \left(1 - \frac{L\,h}{T_0}\right)^{g/(RL)}, \qquad L = 0.0065\ \mathrm{K\,m^{-1}}, \quad \frac{g}{RL} \approx 5.25588, \tag{3}$$

with $g = 9.80665\ \mathrm{m\,s^{-2}}$ and $R = 287.05\ \mathrm{J\,kg^{-1}\,K^{-1}}$ the specific gas constant of dry air. The base $(1 - Lh/T_0)$ is clamped to $[0.1, 2.0]$ and the result to $[100, 1100]$ mbar as a numerical guard.

1.1.2 The Marini mapping form

Both the hydrostatic and wet mappings use the normalised three-parameter continued fraction of Marini, normalised to unity at zenith (`Troposphere::mariniMapping`):

$$\boxed{m(\varepsilon;\, a, b, c) = \cfrac{1 + \cfrac{a}{1 + \cfrac{b}{1 + c}}}{\sin\varepsilon + \cfrac{a}{\sin\varepsilon + \cfrac{b}{\sin\varepsilon + c}}}.} \tag{4}$$

The numerator is the denominator evaluated at $\varepsilon = 90^\circ$ ($\sin\varepsilon = 1$), so $m(90^\circ) = 1$. The hydrostatic and wet mappings, and the different models, differ only in the triple (a, b, c). Two families are implemented.

Niell / GMF coefficients. The default mapping interpolates the Niell tables [5] of (a, b, c) given at five reference latitudes $15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$, each as a mean plus an annually varying amplitude (the coefficient tables inside `Troposphere::getTroposphereMapping`). The seasonal value is

$$v(\phi, t) = \bar{v}(\phi) - \delta v(\phi)\,\cos\frac{2\pi\,(\mathrm{doy} - \mathrm{doy}_0)}{365.25}, \qquad \mathrm{doy}_0 = \begin{cases} 28 & \phi \geq 0 \\ 28 + 182.625 & \phi < 0 \end{cases} \tag{5}$$

(`Troposphere::getTroposphereMapping`, over the day of year returned by `Troposphere::getDayOfYear`). Two details matter. The amplitude term is *subtracted*. The Southern Hemisphere is put in anti-phase by adding half a year to the phase reference, not by changing the sign. The latitude argument is the absolute latitude, linearly interpolated and clamped to $[15^\circ, 75^\circ]$. The wet coefficients have mean values only, with no annual term.

A height correction lifts the hydrostatic mapping from sea level to the station, identical in the Niell and VMF3 paths (`Troposphere::getTroposphereMapping`):

$$m_\mathrm{h}(\varepsilon, h) = m_\mathrm{h}(\varepsilon, 0) + \left[\frac{1}{\sin\varepsilon} - m(\varepsilon;\, a_\mathrm{ht}, b_\mathrm{ht}, c_\mathrm{ht})\right] h_\mathrm{km}, \qquad \begin{aligned} a_\mathrm{ht} &= 2.53 \times 10^{-5}, \\ b_\mathrm{ht} &= 5.49 \times 10^{-3}, \\ c_\mathrm{ht} &= 1.14 \times 10^{-3}, \end{aligned} \tag{6}$$

with the same Marini form (4) evaluated at the height triple. Only the hydrostatic mapping receives this correction.

VMF3 coefficients. When configured for VMF3 [4], the wet/hydrostatic a-coefficients come from the GPT3 grid (next paragraph) and the b, c coefficients are synthesised on the fly by a degree- and order-12 spherical-harmonic expansion of their seasonal columns (`src/core/atmosphere/Vmf3.cpp`). A Cartesian-form associated Legendre recursion builds the real and imaginary solid harmonics V_{nm}, W_{nm} at the station co-latitude and longitude, the $91 = \sum_{n=0}^{12}(n+1)$ harmonic terms are accumulated against the stored `Vmf3Coeffs` tables, and each of $b_\mathrm{h}, c_\mathrm{h}, b_\mathrm{w}, c_\mathrm{w}$ is then evaluated as a five-coefficient mean-plus-annual-plus-semiannual harmonic, with all three steps inside `Vmf3::computeMapping`:

$$v(\mathrm{doy}) = A_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega, \qquad \omega = \frac{2\pi\,\mathrm{doy}}{365.25}. \tag{7}$$

The synthesised (a, b, c) enter the same Marini fraction (4), followed by the height correction (6). If the grid query fails the code falls back to the Niell path without a message, so one bad sample does not stop the run.

1.1.3 GPT3 empirical pressure, temperature and VMF3 a-coefficients

The GPT3 5° grid [4] supplies, per cell, the five harmonic coefficients of pressure, temperature, temperature lapse rate, geoid undulation, and the VMF3 hydrostatic and wet a-coefficients (`Gpt3::load`, in `atmosphere/Gpt3`). Each quantity is the same annual-plus-semiannual harmonic as (7) evaluated per cell (`Gpt3::harmonic`):

$$V(\mathrm{doy}) = a_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega. \tag{8}$$

The grid value sits at the cell's orthometric reference height H_s; it is carried to the station orthometric height $h_\mathrm{ortho} = h - N$ (undulation N) by a barometric pressure reduction and a linear temperature lapse (`Gpt3::query`):

$$P_\mathrm{stn} = P_\mathrm{grid}\big(1 - K_\mathrm{P}\,\Delta h\big)^{n_\mathrm{P}}, \quad T_\mathrm{stn} = T_\mathrm{grid} + \frac{\mathrm{d}T}{\mathrm{d}z}\,\Delta h, \qquad \begin{aligned} K_\mathrm{P} &= 2.26\times10^{-5}\,\mathrm{m}^{-1},\\ n_\mathrm{P} &= 5.225, \end{aligned} \tag{9}$$

with $\Delta h = h_\mathrm{ortho} - H_s$ and $\mathrm{d}T/\mathrm{d}z$ the grid lapse in $\mathrm{K\,km^{-1}}$ applied to Δh in kilometres. The exponent 5.225 is the same $g/(RL)$ ratio as (3), rounded for the grid. Four corners are read and combined by bilinear interpolation in latitude and (wrapped) longitude (also `Gpt3::query`); the resulting P_stn drives the Saastamoinen ZHD of (2), and the interpolated $a_\mathrm{h}, a_\mathrm{w}$ drive the VMF3 mapping.

1.1.4 The gradient mapping and a degenerate fallback

The azimuthal gradient in (1) uses the Chen–Herring mapping [1], coded identically in the Jacobian and the prefit residual (`MeasurementModel::buildJacobianRow`, `StateInitializer::computeRawResidual`):

$$\boxed{m_\mathrm{g}(\varepsilon) = \frac{1}{\sin\varepsilon\tan\varepsilon + 0.0032}, \qquad T^\mathrm{grad} = m_\mathrm{g}(\varepsilon)\big(G_\mathrm{N}\cos\alpha + G_\mathrm{E}\sin\alpha\big).} \tag{10}$$

The partial derivatives $\partial\mathrm{T}/\partial G_\mathrm{N} = m_\mathrm{g}\cos\alpha$ and $\partial\mathrm{T}/\partial G_\mathrm{E} = m_\mathrm{g}\sin\alpha$ are the Jacobian columns of the north and east gradient states (the state vector).

A cold-start guard applies to the *elevation* (hydrostatic and wet) mapping, not to the gradient mapping. When the receiver position is not yet usable (norm below 10^3 m, or non-finite), the

geodetic latitude and height needed by the Niell and VMF3 coefficients are unavailable. The hydrostatic and wet mappings then use the geometry-only Black–Eisner form $1/(\sin\varepsilon + a/(\tan\varepsilon + b))$ (the fallback branch of `Troposphere::getTroposphereMapping`), which keeps the mapping defined during cold start.

1.2 The slant ionospheric delay

The ionosphere is dispersive. To first order it advances the phase and delays the code by the same amount, $40.3\,\mathrm{STEC}/f^2$, where STEC is the slant total electron content in $\mathrm{el\,m^{-2}}$. The uncombined model carries one slant state I_r^i per satellite, referenced to GPS L1, and scales it to each signal by the dispersion ratio.

1.2.1 The $1/f^2$ scaling and its sign

For a signal of frequency f on a state referenced to $f_{\mathrm{ref}} = f_{\mathrm{L1}}^{\mathrm{GPS}}$, the dispersion coefficient is

$$\boxed{\mu_s = \frac{f_{\mathrm{ref}}^2}{f_s^2}, \qquad \frac{\partial P_s}{\partial I} = +\mu_s, \qquad \frac{\partial \Phi_s}{\partial I} = -\mu_s,} \tag{11}$$

so the slant-iono Jacobian column is $+\mu_s$ on code and $-\mu_s$ on phase (`MeasurementModel::buildJacobianRow`). This defines the μ_s of the notation document. Every constellation's slant state is referenced to GPS L1, which lets the between-satellite and between-receiver differencing operators (the differencing document) cancel the receiver iono datum. Because $f_{\mathrm{ref}} = f_{\mathrm{L1}}$, the L1 signal has $\mu = 1$ and the state is the L1 slant delay in metres.

1.2.2 IONEX / GIM thin-shell slant delay

A Global Ionosphere Map gives vertical TEC on a latitude/longitude grid at a single thin shell of height H above a base radius R_E. The slant delay along a ray at elevation ε follows from single-layer geometry (`computeIppGeometry` in `src/core/atmosphere/SphericalHarmonicIonosphere.cpp`, the shared helper that `GlobalIonosphereMap::slantIonoDelay` and the co-estimated spherical-harmonic VTEC model both call so the two never drift). The zenith angle z' at the ionospheric pierce point (IPP) obeys

$$\sin z' = \frac{R_E}{R_E + H}\cos\varepsilon, \qquad \text{mapping } M(\varepsilon) = \frac{1}{\cos z'}, \tag{12}$$

so that $\mathrm{STEC} = M(\varepsilon)\,\mathrm{VTEC}$ (`GlobalIonosphereMap::slantIonoDelay`). The earth-centred angle to the IPP is $\psi = \frac{\pi}{2} - \varepsilon - z'$, and the IPP latitude and longitude come from the spherical-triangle solution along the satellite azimuth α (`computeIppGeometry`). The helper treats the receiver's geodetic latitude ϕ_r as geocentric when crossing the thin shell. The approximation is worth well under a millimetre of L1 delay. The higher-order model (30) later uses the IPP in the same frame:

$$\phi_{\mathrm{IPP}} = \arcsin\big(\sin\phi_r\cos\psi + \cos\phi_r\sin\psi\cos\alpha\big), \tag{13}$$

$$\lambda_{\mathrm{IPP}} = \lambda_r + \arcsin\frac{\sin\psi\,\sin\alpha}{\cos\phi_{\mathrm{IPP}}}. \tag{14}$$

The VTEC at the IPP is interpolated bilinearly in space and, in time, *sun-fixed*: each bracketing map is sampled at the IPP longitude rotated by the Earth's rotation between the map epoch and the query epoch, $\Delta\lambda = (t - t_{\mathrm{map}}) \cdot 360^\circ/86400$ s, before the two are linearly blended

(`GlobalIonosphereMap::interpolateVtec`, over `GlobalIonosphereMap::sampleMap`). The slant delay in metres at frequency f then uses the dispersion constant

$$\boxed{I^{\mathrm{GIM}} = \mathrm{STEC} \cdot \frac{40.3 \times 10^{16}}{f^2} \quad [\mathrm{m}], \qquad \mathrm{STEC} = M(\varepsilon)\,\mathrm{VTEC}\ [\mathrm{TECU}],} \tag{15}$$

the factor 10^{16} converting VTEC from TECU to $\mathrm{el\,m^{-2}}$ and $40.3 = 40.3\ \mathrm{m\,Hz^2\,(el\,m^{-2})^{-1}}$ the first-order refractivity constant (both the constant and the metre conversion live in `GlobalIonosphereMap::slantIonoDelay`).

1.2.3 The GIM iono-weighting pseudo-observation

The GIM is too coarse to constrain the *absolute* slant of one satellite, but it constrains the *relative* slants across a constellation well. The constraint is applied as a Kalman pseudo-observation (`Application::applyGimIonoUpdate`), one row per satellite that owns a slant state above the elevation cutoff. The innovation is the GIM slant minus the current state,

$$z_i = I_i^{\mathrm{GIM}} - I_r^i, \qquad \boldsymbol{H} = \boldsymbol{I}\ \text{(selection of the iono columns)}, \tag{16}$$

with the GIM evaluated at GPS L1 to match the state reference (`Application::applyGimIonoUpdate`). The measurement-noise matrix carries the structure. Per constellation g, with $\mathbf{1}$ the all-ones vector over that constellation's rows,

$$\boxed{\boldsymbol{R} = \bigoplus_{g \in \mathcal{G}} \Big(\mathrm{diag}(\sigma_i^2) + \sigma_c^2\, \mathbf{1}\mathbf{1}^{\mathrm{T}} \Big),} \tag{17}$$

(also `Application::applyGimIonoUpdate`). The rank-one $\sigma_c^2\,\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block gives a very large variance to the per-constellation *common mode* of the innovations. That common mode is the receiver iono datum, an arbitrary offset shared by all satellites of a constellation, which no model can supply. The pseudo-observation therefore leaves it free and constrains only the between-satellite differences, which the GIM does describe. The update uses the same Joseph-form pseudo-update as every other measurement (the Kalman filter document). The iono process noise is documented there (the iono random-walk spectral density).

Why frozen pivots are excluded. In the BSSD parameterisation each constellation's pivot satellite holds its slant-iono state frozen at value 0 with variance 0 as the rank-fixing datum. Such a row is skipped entirely (the frozen-pivot guard in `Application::applyGimIonoUpdate`). A zero-variance state receives zero Kalman gain, so the GIM cannot improve it. It does share the common-mode $\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block of (17) with the other satellites of its constellation. If it were included, its innovation $I_{p_g}^{\mathrm{GIM}} - 0 \sim 5$–$30$ m would enter the gain of every other satellite through the off-diagonal of $\boldsymbol{S}^{-1}$ and shift the relative slants by the pivot's full absolute STEC. Excluding the pivot row is required for a correct result. The common-mode block absorbs the datum offset of the *remaining* rows, so the datum satellite must not appear as a measurement.

1.2.4 The co-estimated spherical-harmonic VTEC field

The network producer can estimate the vertical TEC field itself instead of importing a map. The field is a truncated real spherical-harmonic expansion on the same kind of thin shell, solved jointly with the satellite code biases. The coefficients are *shared* columns of the network normal equations,

registered through `NetworkEstimator::registerVtecCoeff` and reduced onto the shared border like any other network-wide family (the network mode chapter). This section gives the model, its pseudo-observation, and its datum.

The switch is `network.estimate_iono_sh`. It defaults to false, but the standing network configuration arms it and most shipped network configurations set it, so it is the ionosphere model the producer normally runs. Arming it forces `network.estimate_sat_code_bias` on, because the field is what makes the satellite code biases observable. It also suppresses external GIM loading, so the two ionosphere sources are never applied to the same run (`config/GnssConfigLoadNetwork.cpp`; the SH-versus-GIM branch guards the `networkAppendGimIonoRows_` call site in `app/ApplicationNetworkAssembly`).

Basis and normalisation. With β the geographic latitude of the pierce point and s its sun-fixed longitude, the field is

$$\boxed{\mathrm{VTEC}(\beta, s) = \sum_{n=0}^{N} \sum_{m=0}^{n} \bar{P}_{nm}(\sin\beta)\left[a_{nm}\cos(ms) + b_{nm}\sin(ms)\right] \quad [\mathrm{TECU}],} \tag{18}$$

truncated triangularly at degree N (`estimation.global_vtec_sh.max_degree`), which gives $(N{+}1)^2$ real coefficients: one cosine term per $(n, m \geq 0)$ and one sine term per $(n, m \geq 1)$, enumerated once in a fixed order that the shared-column layout mirrors (`SphericalHarmonicIonosphere::coeffs`). The normalisation is

$$\bar{P}_{nm} = N_{nm}\, P_{nm}, \qquad N_{nm} = \sqrt{(2 - \delta_{0m})\,(2n+1)\,\frac{(n-m)!}{(n+m)!}}, \tag{19}$$

chosen so the real surface harmonics $\{\bar{P}_{nm}\cos(ms),\ \bar{P}_{nm}\sin(ms)\}$ are orthonormal under $(1/4\pi)\int(\cdot)\,\mathrm{d}\Omega$ [7]. Because (18) is linear in the coefficients, the basis values are the partial derivatives. `SphericalHarmonicIonosphere::basis` returns $\partial\mathrm{VTEC}/\partial a_{nm}$ and $\partial\mathrm{VTEC}/\partial b_{nm}$ directly, and the Jacobian needs no numerical differentiation. One consequence is used repeatedly below: $\bar{P}_{00} \equiv 1$, so a_{00} is the global mean VTEC.

Frame and geometry. The expansion uses a sun-fixed frame, $s = \lambda_{\mathrm{IPP}} - \lambda_{\odot}$, with $\lambda_{\odot} = \alpha_{\odot} - \mathrm{GMST}$ the sub-solar geographic longitude (`meanSunLongitude`). In this frame the field changes slowly over a session; in a geographic frame it would move westward at the Earth's rotation rate. `estimation.global_vtec_sh.sun_fixed` selects geographic longitude instead. The pierce point and the thin-shell mapping $M(\varepsilon)$ that carries VTEC to STEC come from *the same* `computeIppGeometry` helper that `GlobalIonosphereMap::slantIonoDelay` calls, evaluated at `estimation.global_vtec_sh.layer_height_km` over base radius `earth_radius_km`. Equations (12), (13) and (14) are therefore also the geometry of the co-estimated model. The external map and the co-estimated field share one implementation so that they stay consistent.

The pin row. No measurement observes the field directly. One pseudo-observation per line of sight ties it to the estimated slant states (`Application::networkAppendShIonoRows_`). For every GPS or Galileo LOS above `ionosphere_weighting.elevation_cutoff_deg` whose slant state lives

in that station's local block, the row asserts that the slant state equals the field's prediction:

$$\boxed{\begin{aligned} & g_r^i \equiv I_r^i \;-\; \kappa\, M(\varepsilon) \sum_k \bar{Y}_k(\mathrm{IPP})\, c_k \;=\; 0, \qquad \kappa = \frac{40.3 \times 10^{16}}{f_{\mathrm{L1}}^2}, \\ & \frac{\partial g_r^i}{\partial I_r^i} = +1, \qquad \frac{\partial g_r^i}{\partial c_k} = -\,\kappa\, M(\varepsilon)\, \bar{Y}_k(\mathrm{IPP}). \end{aligned}} \tag{20}$$

Here c_k runs over the field coefficients, $\bar{Y}_k$ over the corresponding basis functions of (18), and κ is the TECU-to-L1-metre conversion already introduced in (15). The $+1$ goes in the station's private block and the $-\kappa M \bar{Y}_k$ terms go in the *shared* border. The second write is what turns an ionosphere prior into a joint estimate. The right-hand side is $-I_r^i$ at the current state, and the shared coefficients enter the absolute solve at zero offset.

The measurement noise on these rows is *the same* common-mode-free per-constellation block as the GIM pseudo-observation, (17), built from the same `ionosphere_weighting.common_mode_sigma_m` over a per-LOS representativeness sigma

$$\sigma_i = \max\big(\texttt{floor_m},\ \sigma_{\mathrm{rep}}\, M(\varepsilon)\, \kappa\big), \qquad \sigma_{\mathrm{rep}} = \texttt{global_vtec_sh.representation_sigma_tecu}, \tag{21}$$

after which the block is whitened by the Cholesky factor of $\boldsymbol{R}$ into unit-information rows before it joins the normal equations. (17) is reused here for a modelling reason. The rank-one $\sigma_c^2\, \mathbf{1}\mathbf{1}^{\mathrm{T}}$ block leaves the iono datum of each (receiver, constellation) pair free. The receiver DCB is part of that common mode, so the field constrains only the *between-satellite* slant structure, which is all a global VTEC surface can describe. As a result a_{00} is only weakly observable, through the spread of $M(\varepsilon)$ across the network, and the datum below has to handle that. GLONASS is excluded from the field (`networkAppendShIonoRows_` skips it). Its FDMA per-slot inter-frequency bias does not cancel in the common-mode block and would leak into the global coefficients.

The Kaula–Tikhonov prior as the estimability datum. The coefficients are shared columns. A coefficient whose pierce-point support has moved over an ocean or polar gap is nearly unobservable in the reduced normal equations and would fail the all-or-nothing rank gate (`NetworkSbasis::assertNullSpaceRank`). A degree-dependent prior is therefore applied to *every* coefficient before the gate runs (`networkInjectShVtecPrior`):

$$\boxed{\begin{aligned} & \sigma_n = \frac{\texttt{kaula_scale_tecu}}{n} \quad (n \geq 1), \qquad \sigma_0 = \texttt{mean_vtec_prior_sigma_tecu}, \\ & \bar{c}_k = \begin{cases} \texttt{mean_vtec_prior_tecu}, & c_k = a_{00}, \\ 0, & \text{otherwise}, \end{cases} \end{aligned}} \tag{22}$$

combined in information space with an optional random-walk anchor toward the previous epoch's estimate, inflated by a per-degree spectral density $q_n = \texttt{q_per_degree}/n^2$ for $n \geq 1$. The prior has two separate roles. The Kaula term provides *estimability*. It keeps every coefficient positive definite, so the declared null-space dimension equals the number of constellations whatever the coverage, and it pulls unobserved coefficients toward climatology each epoch so that they do not random-walk. The a_{00} term is a deliberately *loose* climatology anchor. a_{00} trades against the absolute satellite-DCB level, whose only rigorous datum is the per-(constellation, signal) sum-to-zero constraint of the network mode chapter. Fixing a_{00} to an external mean VTEC would bring back the GIM dependency that co-estimation removes, and would bias every emitted satellite code bias by the difference.

Open extensions: default OFF, not yet field-validated. Two extensions of this model are built, off by default, and still under validation.

(i) A second thin shell (`estimation.global_vtec_sh.layer2_max_degree`, default 0 = single layer) registers a second coefficient set at its own height, and the pin row (20) then sums the shells, each evaluated at *its own* pierce geometry, $\sum_l \kappa\, M_l(\varepsilon)\, \bar{Y}_k(\mathrm{IPP}_l)\, c_{k,l}$ (`ShLayer`, assembled by `shLayers`, which every consumer of the field loops so a shell can never be evaluated at another shell's geometry). The second shell's a_{00} anchor is deliberately *tight*, for the opposite reason to the first shell's. Ground-only geometry barely observes how the mean VTEC divides between the layers: only the difference of the two mapping profiles is sensitive to it. Prior information must therefore hold the split, while the first shell's loose anchor leaves the *total* mean to the data.

(ii) A per-(station, satellite) representativeness deviation (`estimation.global_vtec_sh.rep_error_tau_s`, default 0 = off) replaces the pin-to-zero of (20) with a pin to a tracked first-order Gauss–Markov deviation d, carried in vertical TECU so its level and steady state are elevation-stationary. Writing d^-, P^- for the aged prior, $\hat{u}, \hat{V}$ for the solved posterior mean and variance of the pin functional, and f^2 for the pin-row floor variance, the exact scalar posterior is

$$\hat{d} = d^- + \frac{P^-}{a}\big(\hat{u} - d^-\big), \qquad P^+ = P^-\,\frac{U + f^2}{U + a}, \qquad a = P^- + f^2, \quad U = \frac{\hat{V}\,a}{a - \hat{V}}, \tag{23}$$

with U the data-only information recovered from $\hat{V}$ (`shDevExactUpdate`). Two treatments against ratcheting are built on it, also default OFF.

`rep_error_demean` (`shDevDemean`) removes the information-weighted global and per-station means of the carried priors once per epoch. A deviation component that is common in vertical TECU has row signature $\kappa\, M(\varepsilon)$, the same as the a_{00} column. The pair is therefore exactly quasi-null, and the cycle of harvesting, ageing, and re-asserting the prior ratchets it. The de-mean enforces the state's own contract, which is that it carries per-LOS structure only.

`rep_error_d5_keep` (`shDevD5Retap`) makes the slip response depend on which detector fired. A slip confirmed by a raw detector still *erases* the deviation, following the rule that a stiff time-linked carry must have a wired slip response (the cycle-slip chapter). A re-tap triggered only by the innovation gate keeps the level and resets the variance to the Gauss–Markov steady state.

The warm-LOS counts that arm the de-mean are constants in the seam, not configuration. The loader warns that both treatments are inert unless `rep_error_tau_s` is positive, and that arming the erase-policy split without the de-mean allows the ratchet again.

1.2.5 Klobuchar broadcast seed

The broadcast Klobuchar half-cosine model [3] is used only as a single-frequency a-priori: the cold-start seed of the slant state. It is never used as a dual-frequency correction (`Ionosphere::computeKlobucharDelay`). With elevation and azimuth in semicircles, the earth-centred angle, geomagnetic IPP latitude, local time and obliquity are

$$\psi = \frac{0.0137}{\varepsilon/180 + 0.11} - 0.022, \qquad \phi_i = \mathrm{clip}_{\pm 0.416}\big(\phi/180 + \psi\cos\alpha\big), \tag{24}$$

$$\phi_m = \phi_i + 0.064\cos\big((\lambda_i - 1.617)\pi\big), \qquad F = 1 + 16\,(0.53 - \varepsilon/180)^3, \tag{25}$$

with $\lambda_i = \lambda/180 + \psi\sin\alpha/\cos(\phi_i\pi)$ and local time $t_i = 43200\,\lambda_i + t \pmod{86400}$. Cubic polynomials in the geomagnetic latitude ϕ_m give the amplitude and period,

$$\mathrm{AMP} = \sum_{k=0}^{3} a_k\,\phi_m^k\ (\geq 0), \qquad \mathrm{PER} = \sum_{k=0}^{3} b_k\,\phi_m^k\ (\geq 72000\ \mathrm{s}), \tag{26}$$

and the zenith delay is the truncated half-cosine, mapped to slant by F and scaled to the requested frequency by the f_{L1}^2/f^2 dispersion ratio of (11) (the tail of `Ionosphere::computeKlobucharDelay`):

$$I^{\mathrm{Klob}} = c\,F\Big(5\times10^{-9} + \mathrm{AMP}\left[1 - \tfrac{X^2}{2} + \tfrac{X^4}{24}\right]\Big)\frac{f_{\mathrm{L1}}^2}{f^2}, \quad X = \frac{2\pi(t_i - 50400)}{\mathrm{PER}},\ |X| < 1.57, \tag{27}$$

and the constant 5×10^{-9} s night term alone when $|X| \geq 1.57$. The leading c converts the modelled signal delay from seconds to metres.

1.3 Higher-order ionospheric corrections

The $1/f^2$ scaling covers the first-order term. The second- and third-order delays scale as $1/f^3$ and $1/f^4$ and are applied as deterministic corrections to the modelled range (`HigherOrderIonosphere::slantHigherOrderComponents`). The second order is the magnetic-field–weighted delay

$$\boxed{I_2^P = 7527\,c\,\mathrm{STEC}\,\frac{\boldsymbol{B}\cdot\hat{\boldsymbol{k}}}{f^3}, \qquad I_2^\Phi = -\tfrac{1}{2}\,I_2^P,} \tag{28}$$

where STEC is in $\mathrm{el\,m^{-2}}$, $\boldsymbol{B}$ is the geomagnetic flux density in tesla at the IPP, $\hat{\boldsymbol{k}}$ the receiver→satellite unit vector, and c the speed of light. The constant $7527\,c = e\,A/(2\pi m_e) \approx 2.2564 \times 10^{12}$ combines the electron charge, the first-order refractivity constant A, and the electron mass. The constants and the $-\frac{1}{2}$ phase factor are both in `HigherOrderIonosphere::slantHigherOrderComponents`. The third order depends on the path integral of N_e^2, which a shape factor η relates to the F2-peak density and STEC (also `HigherOrderIonosphere::slantHigherOrderComponents`):

$$\boxed{I_3^P = 2437\,\eta\,N_{\max}\,\frac{\mathrm{STEC}}{f^4}, \qquad N_{\max} = \kappa\,\mathrm{STEC}, \qquad I_3^\Phi = -\tfrac{1}{3}\,I_3^P,} \tag{29}$$

with $\eta = 0.66$ the Hoque–Jakowski shape factor [2], $2437 = 3A^2/8$, and $N_{\max} = \kappa\,\mathrm{STEC}$ the linear Hoque–Jakowski fit ($\kappa = (20-6)\times10^{12}/((4.55-1.38)\times10^{18})$ $\mathrm{el\,m^{-3}/(el\,m^{-2})}$) of the F2-peak electron density to slant STEC. The third-order term is therefore *linear* in $N_{\max}\,\mathrm{STEC}$, not in STEC^2. Both corrections are below a millimetre at L1. They differ by satellite, so between-satellite differencing does not remove them, and each is applied with its own sign on code and on phase.

1.3.1 The geomagnetic dipole field

The field at the IPP is a centred, tilted magnetic dipole evaluated in ECEF (`MagneticField::fieldAtEcef`):

$$\boxed{\boldsymbol{B}(\boldsymbol{r}) = B_0\left(\frac{R_E}{r}\right)^3\big(\hat{\boldsymbol{m}} - 3(\hat{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\,\hat{\boldsymbol{r}}\big),} \tag{30}$$

with $r = \|\boldsymbol{r}\|$, $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$, $B_0 = 30\times10^{-6}$ T the equatorial surface field, $R_E = 6371.2$ km the IGRF reference radius, and $\hat{\boldsymbol{m}}$ the unit axis pointing to the geographic location of the geomagnetic north pole, IGRF-13 epoch 2020.0 at $(80.65°, -72.68°)$ (`MagneticField::dipoleAxisEcef`). The Arctic geomagnetic pole is the dipole's *south* pole. With $\hat{\boldsymbol{m}}$ pointing at it, (30) makes $\boldsymbol{B}$ point horizontally toward magnetic north at the magnetic equator, as a compass does. This convention sets the sign of $\boldsymbol{B}\cdot\hat{\boldsymbol{k}}$ and therefore of I_2 in (28). IONEX is a thin-shell model, so the IPP ECEF position

is rebuilt from the geocentric IPP latitude and longitude at the IONEX base-plus-shell radius (`HigherOrderIonosphere::slantHigherOrderComponents`).

The a-priori ZHD, the estimated ZWD and gradient states, the uncombined slant-iono state, and these corrections account for every atmospheric term of the undifferenced observation equation. The differencing document describes the operators that then remove the receiver-common parts.

References

[1] Guang Chen and Thomas A. Herring. "Effects of atmospheric azimuthal asymmetry on the analysis of space geodetic data". In: *Journal of Geophysical Research: Solid Earth* 102.B9 (1997), pp. 20489–20502. DOI: 10.1029/97JB01739.

[2] M. Mainul Hoque and Norbert Jakowski. "Estimate of higher order ionospheric errors in GNSS positioning". In: *GPS Solutions* 12.2 (2008), pp. 87–97. DOI: 10.1007/s10291-007-0069-5.

[3] J. A. Klobuchar. "Ionospheric time-delay algorithm for single-frequency GPS users". In: *IEEE Transactions on Aerospace and Electronic Systems* AES-23.3 (1987), pp. 325–331. DOI: 10.1109/TAES.1987.310829.

[4] Daniel Landskron and Johannes Boehm. "VMF3/GPT3: refined discrete and empirical troposphere mapping functions". In: *Journal of Geodesy* 92.4 (2018), pp. 349–360. DOI: 10.1007/s00190-017-1066-2.

[5] A. E. Niell. "Global mapping functions for the atmosphere delay at radio wavelengths". In: *Journal of Geophysical Research: Solid Earth* 101.B2 (1996), pp. 3227–3246. DOI: 10.1029/95JB03048.

[6] J. Saastamoinen. "Atmospheric correction for the troposphere and stratosphere in radio ranging of satellites". In: *The Use of Artificial Satellites for Geodesy*. Ed. by S. W. Henriksen, A. Mancini, and B. H. Chovitz. Vol. 15. Geophysical Monograph Series. Washington, D.C.: American Geophysical Union, 1972, pp. 247–251. DOI: 10.1029/GM015p0247.

[7] J. Sanz Subirana, J. M. Juan Zornoza, and M. Hernández-Pajares. *GNSS Data Processing, Volume I: Fundamentals and Algorithms*. ESA TM-23/I. Noordwijk, The Netherlands: ESA Communications, 2013.